

Distance-Triggered Switching Between LLM and Bayesian Optimization: A Primal–Dual Optimization Framework

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1 Setup and Notation

Let $t \in \{1, \dots, T\}$ index discrete search steps with finite horizon T . At each step, a Large Language Model (LLM) and a Bayesian Optimizer (BO) each propose one candidate, inducing empirical distributions over the search space $\mathcal{X}$.

Empirical distributions.

$$\begin{aligned} P_t &= \text{empirical distribution of LLM samples up to step } t, \\ Q_t &= \text{empirical distribution of BO samples up to step } t. \end{aligned}$$

Generic discrepancy metric. Let $d(P_t, Q_t)$ denote any valid discrepancy between P_t and Q_t . Candidate choices include:

- **Wasserstein-2** (W_2): metrizes weak convergence; connects to optimal transport.
- **Maximum Mean Discrepancy** (MMD): kernel-based, $O(n^2)$, no sorting required.
- **KL Divergence**: information-theoretic; requires density estimation.
- **Energy Distance**: distribution-free; $O(n^2)$.

Expected distance. Since both trajectories are stochastic, define the *expected distance* at step t :

$$D(t) = \mathbb{E}[d(P_t, Q_t)]. \quad (1)$$

In practice, estimated over K independent runs:

$$\hat{D}(t) = \frac{1}{K} \sum_{k=1}^K d(P_t^{(k)}, Q_t^{(k)}). \quad (2)$$

Normalized distance. To remove metric-specific scaling and obtain a universal $[0, 1]$ signal, define:

$$\tilde{D}(t) = \frac{D(t) - D(\infty)}{D(0) - D(\infty)}, \quad \tilde{D}(0) = 1, \quad \tilde{D}(\infty) \rightarrow 0. \quad (3)$$

[Convex decrease] $\tilde{D}(t)$ is convex and decreasing in t : fast initial drop followed by flattening. The *elbow* is the point of maximum curvature.

Cost parameters. Let $c_{\text{LLM}} > c_{\text{BO}} > 0$ denote the per-step costs (time or money) of LLM and BO queries respectively, and let $\Delta c = c_{\text{LLM}} - c_{\text{BO}} > 0$.

2 Primal Problem

We choose a *switching step* $\tau \in [0, T]$ at which the search process transitions from LLM to BO. The objective is to maximize log-accuracy subject to a *belief-closeness* constraint (the LLM must be “on track” before handing off) and a *budget* constraint.

2.1 Objective

Define total log-accuracy as:

$$F(\tau) = \underbrace{\log A_{\text{LLM}}(\tau)}_{\text{LLM contribution}} + \underbrace{\log A_{\text{BO}}(T - \tau)}_{\text{BO contribution}} + \underbrace{\gamma \phi(\tilde{D}(\tau))}_{\text{synergy}}, \quad (4)$$

where:

- $\log A_{\text{LLM}}(\tau)$: log-accuracy accrued by LLM over τ steps — concave increasing (diminishing returns).
- $\log A_{\text{BO}}(T - \tau)$: log-accuracy of BO given $T - \tau$ steps with a warm start from LLM — concave in $T - \tau$, and *improving* in τ (more LLM exploration $\Rightarrow$ better prior for BO).
- $\phi(\tilde{D}(\tau))$: synergy term, *decreasing* in $\tilde{D}$ — smaller distance implies higher-quality handoff.
- $\gamma \geq 0$: synergy weight.

2.2 Constraints

- **Closeness constraint**: the LLM and BO beliefs must be sufficiently aligned at the switching point:

$$g_1(\tau) := \tilde{D}(\tau) - \epsilon \leq 0, \quad (5)$$

for a threshold $\epsilon > 0$.

- **Budget constraint**: total cost must not exceed budget B :

$$g_2(\tau) := \tau c_{\text{LLM}} + (T - \tau) c_{\text{BO}} - B = \tau \Delta c - B' \leq 0, \quad (6)$$

where $B' = B - T c_{\text{BO}}$, so the budget constraint simplifies to $\tau \leq \bar{\tau} := B' / \Delta c$.

2.3 Primal Formulation

$$\boxed{\begin{array}{ll} \max_{\tau \in [0, T]} & F(\tau) = \log A_{\text{LLM}}(\tau) + \log A_{\text{BO}}(T - \tau) + \gamma \phi(\tilde{D}(\tau)) \\ \text{s.t.} & g_1(\tau) := \tilde{D}(\tau) - \epsilon \leq 0 \quad [\text{closeness}] \\ & g_2(\tau) := \tau \Delta c - B' \leq 0 \quad [\text{budget}] \\ & 0 \leq \tau \leq T. \end{array}} \quad (7)$$

Remark 1 (Convexity). $F(\tau)$ is concave (sum of concave terms). Both g_1 and g_2 are convex in τ ($\tilde{D}$ convex decreasing; g_2 linear). Hence (7) is a convex program, and Slater's condition holds generically.

3 Lagrangian and Dual Problem

3.1 Lagrangian

Attach multiplier $\mu \geq 0$ to the closeness constraint and $\lambda \geq 0$ to the budget constraint:

$$\mathcal{L}(\tau, \mu, \lambda) = F(\tau) - \mu(\tilde{D}(\tau) - \epsilon) - \lambda(\tau \Delta c - B'). \quad (8)$$

Expanding:

$$\mathcal{L}(\tau, \mu, \lambda) = \log A_{\text{LLM}}(\tau) + \log A_{\text{BO}}(T - \tau) + (\gamma\phi - \mu)\tilde{D}(\tau) - \lambda\tau\Delta c + \mu\epsilon + \lambda B'. \quad (9)$$

3.2 Dual Function and Dual Problem

The dual function is:

$$g(\mu, \lambda) = \max_{\tau \in [0, T]} \mathcal{L}(\tau, \mu, \lambda). \quad (10)$$

The **dual problem** is:

$$\boxed{\min_{\mu, \lambda \geq 0} g(\mu, \lambda) = \min_{\mu, \lambda \geq 0} \max_{\tau \in [0, T]} \mathcal{L}(\tau, \mu, \lambda).} \quad (11)$$

Theorem 1 (Strong Duality). *Under Slater’s condition (primal is strictly feasible) and concavity of F , strong duality holds:*

$$\max_{\tau} \min_{\mu, \lambda \geq 0} \mathcal{L} = \min_{\mu, \lambda \geq 0} \max_{\tau} \mathcal{L}. \quad (12)$$

There is no duality gap, and primal and dual solutions coincide.

4 KKT Conditions

At the saddle point $(\tau^*, \mu^*, \lambda^*)$, the following conditions are necessary and sufficient.

4.1 Stationarity

Setting $\partial \mathcal{L} / \partial \tau = 0$:

$$\left. \frac{d \log A_{\text{LLM}}}{d\tau} \right|_{\tau^*} - \left. \frac{d \log A_{\text{BO}}}{d\tau} \right|_{\tau^*} + \gamma \phi'(\tilde{D}(\tau^*)) \tilde{D}'(\tau^*) = \mu^* \tilde{D}'(\tau^*) + \lambda^* \Delta c. \quad (13)$$

Rearranging into the canonical form:

$$\boxed{\underbrace{\frac{d \log A_{\text{LLM}}}{d\tau} - \frac{d \log A_{\text{BO}}}{d\tau}}_{\text{marginal accuracy gain: LLM over BO}} = \underbrace{(\mu^* - \gamma \phi'(\tilde{D})) \tilde{D}'(\tau^*)}_{\text{net cost of belief misalignment}} + \underbrace{\lambda^* \Delta c}_{\text{budget cost of one more LLM step}}} \quad (14)$$

4.2 Primal and Dual Feasibility

$$\tilde{D}(\tau^*) \leq \epsilon, \quad \tau^* \Delta c \leq B', \quad (15)$$

$$\mu^* \geq 0, \quad \lambda^* \geq 0. \quad (16)$$

4.3 Complementary Slackness

$$\mu^* (\tilde{D}(\tau^*) - \epsilon) = 0, \quad (17)$$

$$\lambda^* (\tau^* \Delta c - B') = 0. \quad (18)$$

5 Interpretation

5.1 Stationarity: The Fundamental Trade-off

Equation (14) states that at the optimal switching point τ^* , the *marginal benefit of one more LLM step* (left-hand side) must exactly equal the combined cost of: (i) the remaining belief misalignment penalized at rate μ^* , and (ii) the budget premium of LLM over BO penalized at rate $\lambda^* \Delta c$. This is the fundamental switching trade-off.

5.2 Complementary Slackness: Two Regimes

Case 1: $\mu^* = 0$ (closeness constraint slack). The closeness threshold ϵ is non-binding: $\tilde{D}(\tau^*) < \epsilon$. The accuracy objective alone determines τ^* . Belief alignment is achieved automatically before the budget is exhausted.

Case 2: $\mu^* > 0$ (closeness constraint binding). The switch occurs *exactly* at the elbow:

$$\tilde{D}(\tau^*) = \epsilon. \quad (19)$$

This is consistent with the empirical observation that all metrics coincide at $t^* = 5$. Here μ^* is the **shadow price of belief alignment**: it quantifies how much additional accuracy would be gained by tolerating a marginally looser threshold, i.e., $\partial g / \partial \epsilon = -\mu^*$.

5.3 Dual Variables as Prices

6 Time–Distance Bijection

Definition 1 (First hitting time). *For threshold $\epsilon > 0$, define:*

$$\tau^*(\epsilon) = \inf \{t \geq 0 : \tilde{D}(t) \leq \epsilon\}. \quad (20)$$

Definition 2 (Implied threshold). *For the accuracy-optimal switching step τ_{acc}^* (solution of the unconstrained version of (7)), define:*

$$\epsilon^*(\tau) = \tilde{D}(\tau_{\text{acc}}^*). \quad (21)$$

Variable	Primal Meaning	Dual Interpretation
τ^*	Optimal switching step	—
ϵ	Closeness threshold	Tightness priced by μ^*
μ^*	—	$\partial g/\partial \epsilon = -\mu^*$: value of relaxing closeness
λ^*	—	$\partial g/\partial B = -\lambda^*$: value of relaxing budget

Table 1: Primal and dual variable interpretations.

Proposition 1 (Equivalence). *The primal problem (optimize over τ) and the threshold problem (find ϵ such that the first hitting time equals the accuracy-optimal step) are equivalent:*

$$\tau^*(\epsilon^*) = \tau_{\text{acc}}^*, \quad (22)$$

related by the monotone map $\tilde{D}(t)$. Under strong duality, μ^* measures how tightly the two representations are coupled.

Metric invariance. Empirically, W_2 , MMD, KL, and Energy distance all exhibit their elbow at the same step $t^* = 5$, with metric-specific threshold values:

$$d_{W_2}(t^*) \approx 1.0, \quad d_{\text{MMD}}(t^*) \approx 0.8, \quad d_{\text{KL}}(t^*) \approx 0.2, \quad d_{\text{E}}(t^*) \approx 0.7. \quad (23)$$

After normalization via (3), all metrics yield the same $\tilde{\epsilon}^*$, confirming that:

1. The switching point τ^* is *metric-agnostic*.
2. Metric choice affects only the scale of ϵ^* , not the decision.
3. Consistency across metrics is a **robustness certificate** for the elbow as a genuine structural feature of LLM-to-BO convergence.

7 Handling Stochasticity via Expectation

Since both trajectories are stochastic, all distance evaluations are replaced by their expectations. The primal problem (7) becomes:

$$\begin{array}{ll} \max_{\tau \in [0, T]} & \mathbb{E}[F(\tau)] \\ \text{s.t.} & \mathbb{E}[\tilde{d}(\tau)] \leq \epsilon \\ & \mathbb{E}[\text{cost}(\tau)] \leq B \\ & 0 \leq \tau \leq T, \end{array} \quad (24)$$

where $\mathbb{E}[\tilde{d}(\tau)]$ is estimated by $\hat{D}(\tau)$ from (2).

Remark 2. *The expectation operator preserves convexity of $\tilde{D}(t)$ (expectation of a convex function is convex). Hence the full framework — Lagrangian, KKT conditions, and strong duality — carries through unchanged to the stochastic setting.*

Remark 3 (Practical estimation). *In practice, K independent runs yield $\hat{D}(t)$ with confidence intervals. As $K \rightarrow \infty$, $\hat{D}(t) \rightarrow D(t)$ uniformly, and the estimated switching point $\hat{\tau}^*$ concentrates around the true τ^* .*

8 Summary

Component	Primal View	Dual View
Decision variable	Choose τ^*	Choose $\tilde{\epsilon}^*$
Core constraint	$\tilde{D}(\tau) \leq \epsilon$	First hitting time = τ^*
Shadow price μ^*	Cost of enforcing closeness	Value of tighter belief alignment
Shadow price λ^*	Cost of budget constraint	Value of additional budget
Elbow at $t = 5$	Optimal switching step	Binding distance threshold
Metric agreement	Robustness of τ^*	$\tilde{\epsilon}^*$ metric-specific; τ^* is not

Table 2: Summary of primal and dual structure.

The central result is that **optimizing the switching time** and **finding the distance threshold** are dual representations of the same problem, linked by the normalized distance curve $\tilde{D}(t)$ and made exact by strong duality. The dual variables μ^* and λ^* have natural economic interpretations as the shadow prices of belief alignment and budget, respectively.